

Assistive Backpack for Independent Living

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The Problem

Individuals reminded they need to go somewhere but not what they need to bring

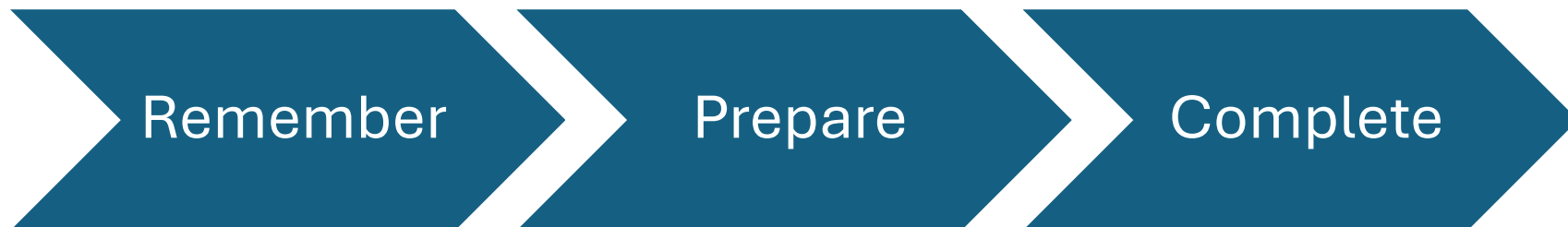
Backpack detects when items are placed into it and alerts the user what else is needed before leaving

Backpack has scheduled alerts and custom routines to match users needs

Maintain independence from carer to feel more control over daily life

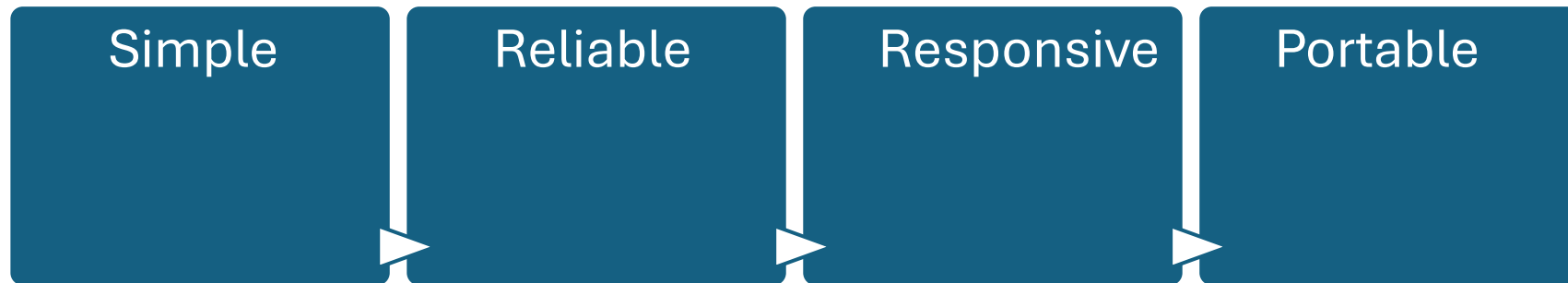
Why this Matters

- 2.5 billion people globally need at least one assistive product, with around 10% of older adults diagnosed with dementia
- 1.7 million Australians had psychosocial disability in 2022
- 35.6% of Australians with psychosocial disability experience memory problems restricting everyday tasks
- "You have an appointment" to "Do you have what you need for this appointment?"



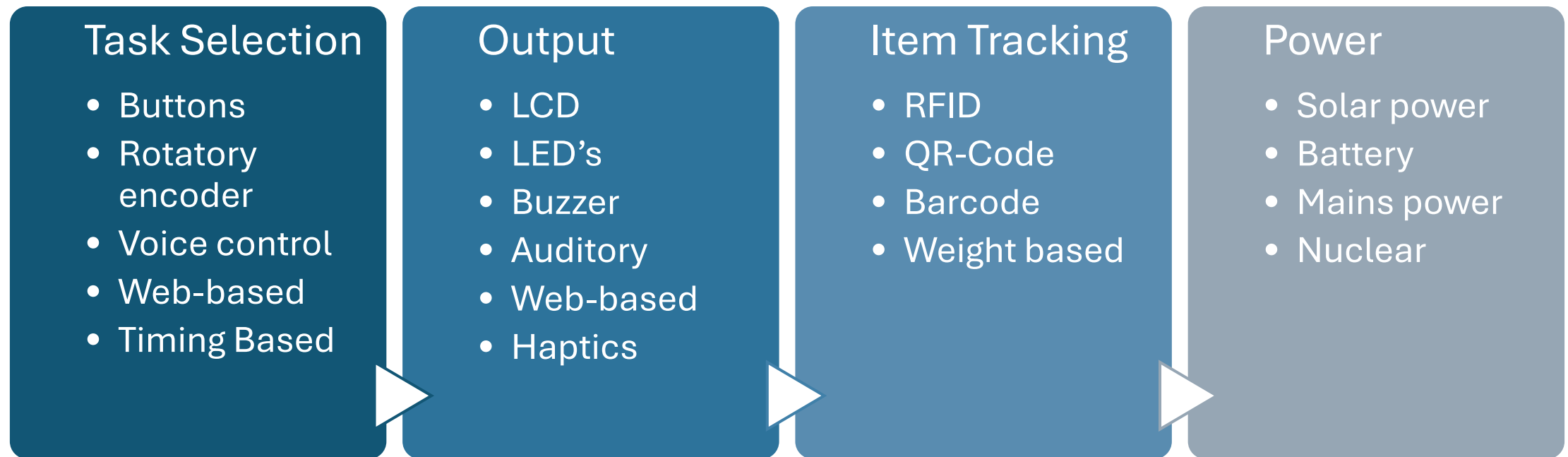
From Problem to Requirements

User need	Engineering Requirement
Know whether items are packed	>95% RFID detection accuracy
Receive quick feedback	<2s warning/checklist response
Simplistic use	Task selected in <5 button presses
Immediate interaction	<200ms button response time
Suitable for a full day	>8h battery life



Requirements to Implementation

- Many possible solutions **brainstormed** for key subsystems.
- Solutions evaluated for **ease of use**, longevity and **low power usage**.



Morphological Design Process

Technical Design

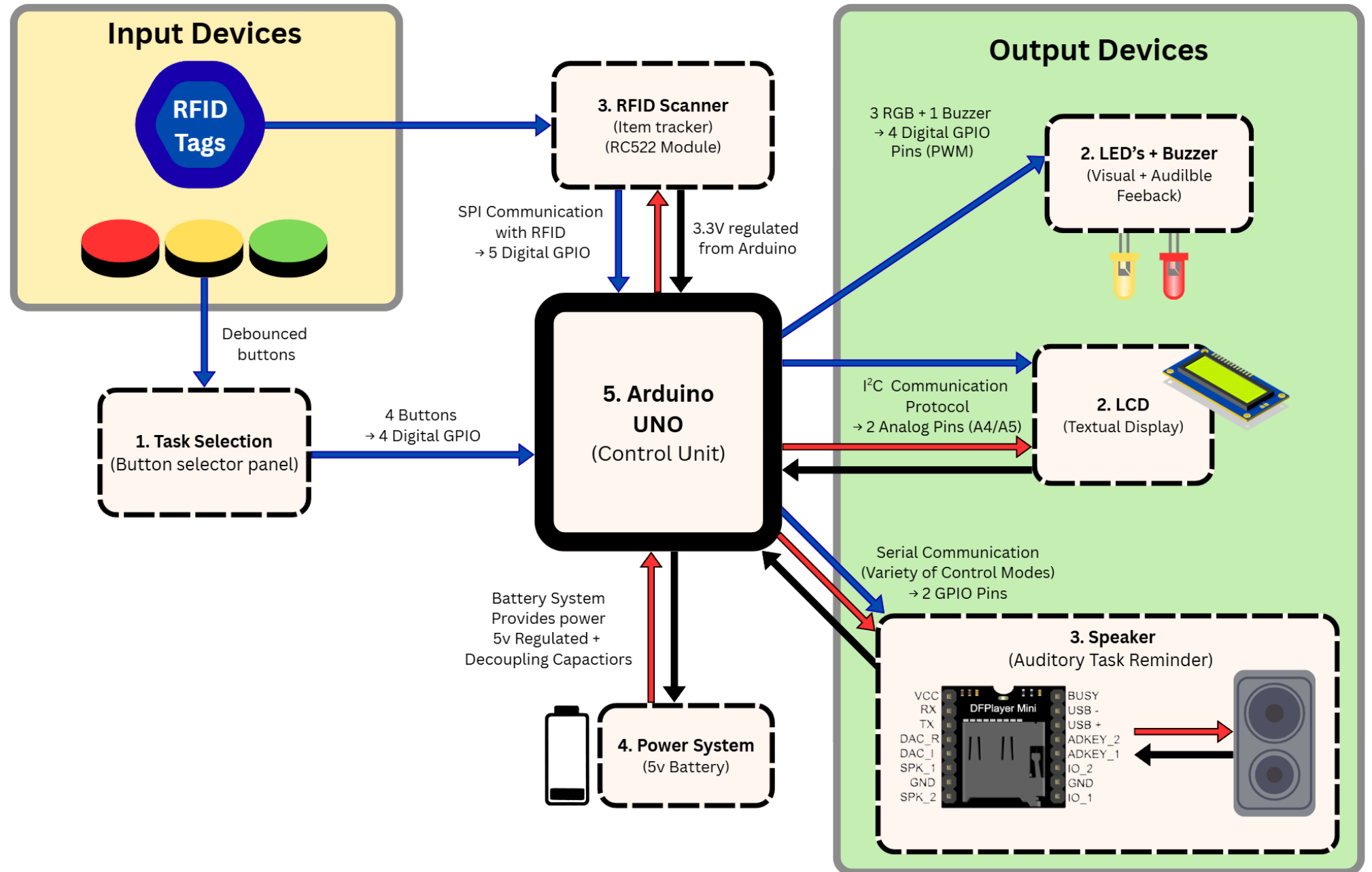


Figure 1 : High Level Project Architecture

Power Calculations

- The backpack system will run on a **single battery pack**.
- Maximum **current draw** and 8 hr **uptime** requirement was used to estimate battery capacity
- **5v 2700mAh battery** was found suitable for this project

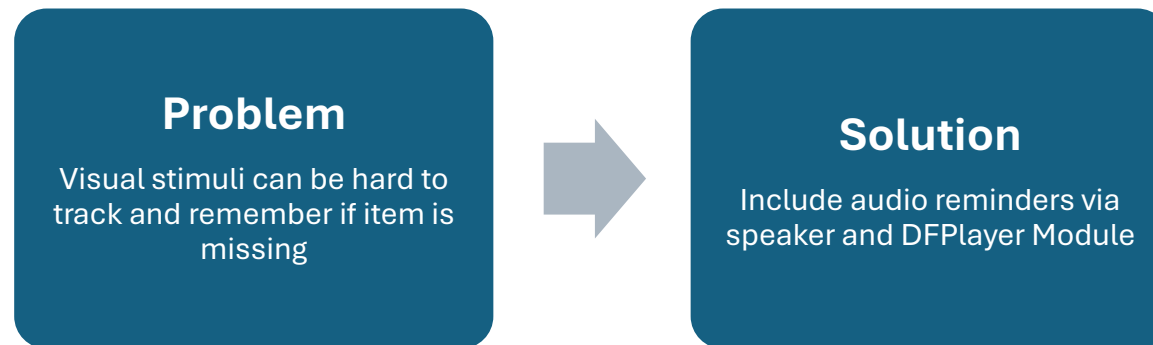
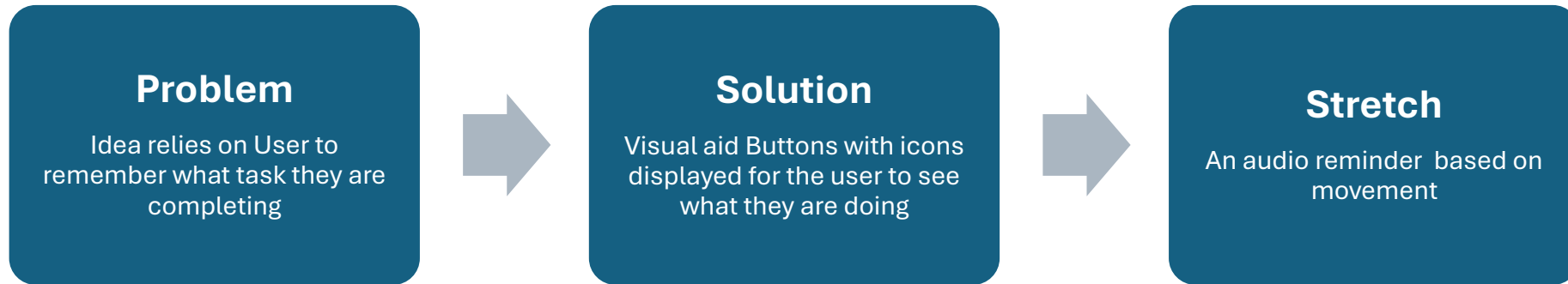
Device	Max Current
Arduino UNO	50mA
RGB LED's *2	60mA
16x2 LCD	50mA
RC522 RFID Module	20mA
Buzzer	20mA
DFMini MP3 Player	20mA
3W Speaker (running at 5V)	600mA
Total	820mA

$$C_{\text{battery}} = \frac{I_{\text{cont}} \times \text{Up Time}}{0.8} + \frac{I_{\text{speaker}} \times \text{On time}}{0.8}$$

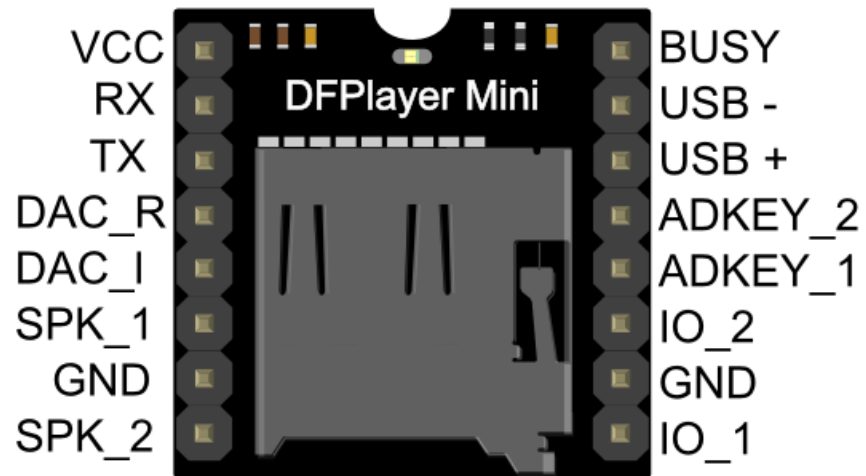
$$C_{\text{battery}} = \frac{220 \times 8\text{hrs}}{80\% \text{ Charge}} + \frac{600 \times 0.67\text{hrs}}{80\% \text{ Charge}}$$

$$C_{\text{battery}} = 2700\text{mAh}$$

Design Updates



DFPlayer Mini



Benefits:

- More accessibility access
- Does not require external frequency adjustment
- Volume and EQ adjustment
- Ability to store thousands of mp3 files

The Risks

Risk	Impact	Mitigation
RFID missed detection	High	Repeated testing and optimising tag placement
Battery Life Below Target	High	Measure Current and Optimise Power
Bag is hard to identify for user	High	Use brightly coloured bag to help identify
Team member doesn't finish part on time	High	Clear deadlines set and recorded. Messaging available on two platforms and regular team meetings.
UI too complex	Medium	User Testing + Introducing New Audio System

Assumptions

User understands icons

User will remember to
charge and take backpack

Items will have RFID
attached

User can read or hear the
reminders

Q&A

Why RFID over Weight Sensor?

- Weight could identify has been added or removed
- RFID is item-specific identifier

Smartphone?

- Adds dependencies
- Less simplistic for memory-challenged patients
- Keeping In Touch Everyday (KITE) found that smart devices should be disguised and integrated better for daily routines

Item Not Detected

- Checklist is incomplete
- New audio module (DFPlayer) reminds user

Why 8 Hours?

- Covers a standard work day
- 10 hours is nominal

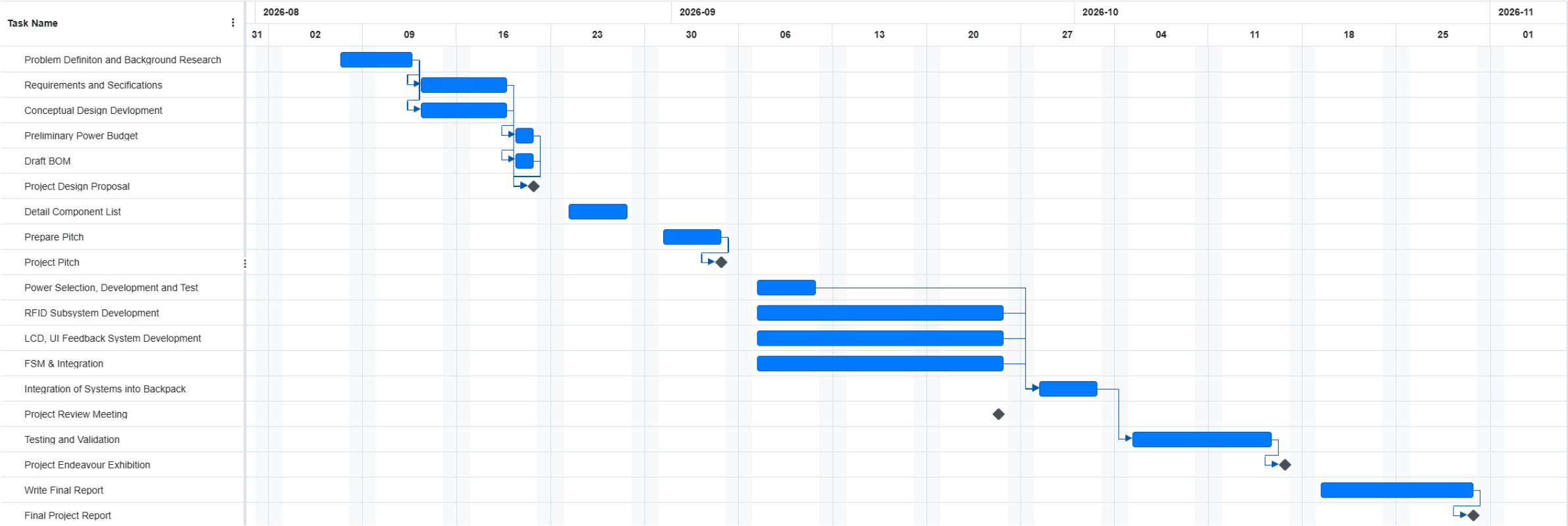
RFID Limitations

- Has not been tested yet
- RFID reader will have checkpoint with chime rather than placing straight in bag

Buttons over Voice Activation?

- Cost
- Reliability
- Power constraints

Gantt Chart for project timeline

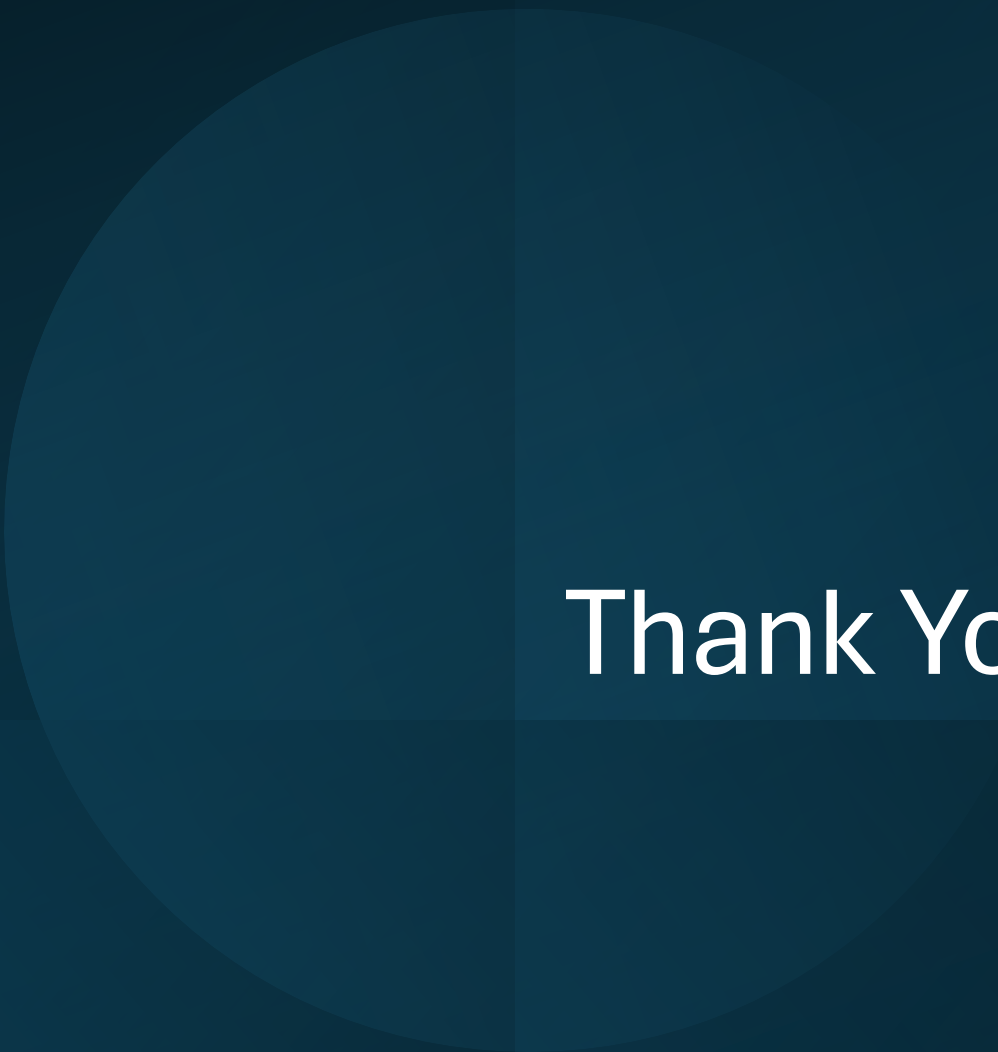


Task	Emerson	Callum	Oliver
Requirements & documentation	R	C	C
RFID / item-detection subsystem	R	C	I
LCD, UI & feedback subsystem	I	R	C
Firmware (FSM & integration)	I	I	R
Power system & battery	C	I	R
Enclosure & backpack build	R	C	I
Testing & validation	C	R	C
Final report & presentation	C	R	C

R: Responsible

C: Consulting

I: Informed



Thank You